MILITARY SPECIFICATION

BOOSTER, FUZE, MARK 60 MOD 3

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the MARK 60 MOD 3 Fuze Booster, referred to herein as "the booster."

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification for.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733-1500, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 1336

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MIL-T-18303	Test Procedures; Preproduction, Acceptance, and Life for Aircraft Electronic Equipment, Format for.
MIL-C-21723	Composition CH-6.

STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-453	Inspection, Radiographic.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-1695	Environments, Working, Minimum Standards for.
MIL-STD-45662	Calibration Systems Requirements.

2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DOCUMENTS

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171-178	Transportation.

DEPARTMENT OF DEFENSE (DOD)

DOD 4145.26M	DOD Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material.

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DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

67A84D518	Cap, Thermal, Insulating.
PBL 1137AS2190	Booster, Fuze, MARK 60 MOD 3.
SA 1137AS2195	Fixture, Cook-off Test.
LD 480141	Lead, Explosive, MK 9, MOD 0.
SA 2747612	Fixture, Vibration, Three Plane.
SA 2747613	Fixture, Jolt Test.
SA 2747614	Fixture, 40-Foot Drop.

(Copies of specifications, standards, drawings, and other Government documents, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A-108	Steel Bars, Carbon, Cold-finished, Standard Quality, Specification for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be avaliable in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Item description. The booster initiates the warhead of a guided missile and is composed of a housing, cap, plastic cup, snap ring, connector, wave spring, identification plate, and an explosive pellet containing approximately 84 grams of Composition CH-6 explosive (see PBL 1137AS2190).

3.1.1 Government-furnished property. Unless otherwise specified in the contract or purchase order (see 6.2.1), the Government will furnish the contractor with Composition CH-6 explosive in accordance with MIL-C-21723, in a quantity sufficient to meet contract requirements.

3.2 Characteristics.

3.2.1 Function and output. The booster shall function on the first and only trial when detonated with an exploding bridge wire (EBW) detonator and a MK 9 MOD 0 lead as shown on Figures 1 and 2. The output shall produce a depth of dent of not less than .140 inch in the steel block as shown on Figure 3.

3.2.2 Radiographic. The booster shall meet the following requirements:

a. There shall be no evidence of loose powder or inclusion of foreign material.

b. There shall be no evidence of internal voids, fissures or cracks or low density areas in the explosive pellet.

c. All parts shall be properly located as shown on PBL 1137AS2190.

3.2.3 Environmental conditions. The booster shall not be damaged nor shall safety or subsequent performance be degraded by exposure to the environments specified in 3.2.3.1 through 3.2.3.3.

3.2.3.1 Transportation vibration-temperature. The booster shall show no evidence of cracks, dents, ruptures, or other deformities, and shall function after vibration testing at ambient temperature in all three axes at the following levels:

Frequency (Hz)	Test Level
5-11	10 mm displacement
11-37	2.5 g acceleration, peak
37-52	0.9 mm displacement
52-500	5.0 g acceleration, peak

3.2.3.2 Temperature and humidity. The booster shall show no evidence of cracks, dents, ruptures, or other deformities, and shall function after exposure to 28 days of temperature/humidity cycling ranging from +160°F and 95 percent humidity to -65°F.

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3.2.3.3 Random vibration. The booster shall show no evidence of cracks, dents, ruptures, or other deformities, and shall function after exposure to the vibration profile with the parameters f_1 = 1000 Hz, f_2 = 2000 Hz, w_1 = .01 g^2/Hz, and w_2 = .12 g^2/Hz in each of the three perpendicular axes at temperatures of +160°F and -65°F for 60 minutes per axis.

3.2.4 Safety conditions.

3.2.4.1 Twelve meter drop. The booster shall not detonate nor burn (but need not be operable), and shall be safe to handle and dispose of after being dropped one time from a height of 12 meters onto a steel plate setting on a concrete base.

3.2.4.2 Jolt. The booster shall show no evidence of cracks, dents, ruptures, or other deformities, and shall be safe to handle and dispose of after exposure to 1750 drops of 4 inches onto a padded anvil at a rate 35 drops per minute in each of three different orientations.

3.2.4.3 Cook-off. The booster shall not detonate while exposed to a flame temperature of 1500°F. Any booster reaction more severe than burning shall not be acceptable.

3.3 First article. When specified in the contract or purchase order a sample shall be subjected to first article inspections (see 4.2 and 6.3).

3.4 Design and construction.

3.4.1 Production drawings. The booster shall meet all the requirements of and shall be manufactured and assembled in accordance with the drawings and documents listed on PBL 1137AS2190.

3.4.2 Safety precautions. As a minimum, the safety precautions of DOD 4145.26M shall apply during manufacture, handling, storage, transportation, and inspection of the booster, but shall not be considered as a substitution for the contractor's own safety procedures developed to accommodate the hazards associated with the booster.

3.4.3 Standards of manufacture.

3.4.3.1 Product cleanliness. No oils or solid lubricants of any kind shall be used on any part of the booster after final cleaning of components, except as specified on the drawings or herein.

3.4.3.2 Working environment. The standards of the working environment shall meet the levels and requirements of MIL-STD-1695.

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3.4.3.3 Workmanship. Workmanship of the booster shall meet the requirements of MIL-A-2550. The booster shall show no evidence of external cracks, burrs, dents, loose powder, or other deformities that may interfere with functioning or fit at the next assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the contracting activity. The contracting activity reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All boosters must meet all requirements of Sections 3 and 5 (see Table I). The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the contracting activity for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the contracting activity to acceptance of defective material.

4.1.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.2).

b. Quality conformance inspections (see 4.3).

4.2 First article inspections. Unless otherwise specified in the contract or purchase order (see 6.3), after award of contract and prior to production the contractor shall perform the first article inspections shown on Figure 4. The contracting activity reserves the right to witness all tests, and the contractor shall notify the contracting activity 15 days prior to testing. As specified in the contract or purchase order, first article inspections shall be performed on 30 boosters that have been produced with the equipment, material, and procedures to be used in production. First article approval is valid only on the contract under which it is granted unless extended by the contracting activity to other contracts. Any production prior to approval of the first article shall be at the risk of the contractor.

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TABLE I. Requirement verification index.

Requirement title	Requirement paragraph no.	Verification paragraph no.
CHARACTERISTICS		
Function and output	3.2.1	4.6.1
Radiographic	3.2.2	4.6.2
ENVIRONMENTAL		
Transporation vibration - temperature	3.2.3.1	4.6.3.1
Temperature and humidity	3.2.3.2	4.6.3.2
Random vibration	3.2.3.3	4.6.3.3
SAFETY CONDITIONS		
Twelve-meter drop	3.2.4.1	4.6.4.1
Jolt	3.2.4.2	4.6.4.2
Cook-off	3.2.4.3	4.6.4.3
DESIGN AND CONSTRUCTION		
Production drawings	3.4.1	4.6.5
Safety precautions	3.4.2	4.6.5
Standards of manufacture	3.4.3	4.6.5
PACKAGING		
Preservation and packaging	5.1	4.6.6
Packing	5.2	4.6.6
Marking	5.3	4.6.6

4.2.1 First article inspection sequence. The first article of 30 boosters shall be subjected to the inspections shown on Figure 4 in the sequence shown. Samples for hot temperature testing (+160°F) and cold temperature testing (-65°F) must be identified prior to test and the allocation retained throughout the test sequence. At the conclusion of the inspection sequence and after receipt of first article approval, any remains of the first article shall be disposed of at the discretion of the cognizant demolition officer.

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4.2.2 First article acceptance and rejection criteria. Each first article booster shall meet the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections shown on Figure 4, the applicable performance requirements of Section 3 shall be met on the first and only trial; one or more failures shall be cause for refusal to grant first article approval. For the function and output test of 4.6.1, a depth of dent of less than 0.140 inch from any first article booster shall be cause for rejection of the first article. When computed in accordance with 4.6.1.3d, the first article shall be accepted if the quantity (X-L)/s is equal to or greater than MIL-STD-414, Table B-1, Sample size code letter G, k value, Acceptable Quality Level (AQL) of 0.65 percent defective. A quantity less than the k value shall be cause for rejection of the first article.

4.3 Quality conformance inspections.

4.3.1 Inspection lot. Unless otherwise specified in the contract or purchase order (see 6.2.1), an inspection lot shall consist of completed boosters, fabricated under the same conditions and offered for acceptance inspection at the same time.

4.3.1.1 Inspection lot size. The inspection lot size shall be as specified in the contract or purchase order (see 6.2.1).

4.3.1.2 Inspection lot acceptance inspection sequence. After the inspection lot has been formed and identified by lot number, the lot shall be subjected to the inspections shown on Figure 5 in the sequence shown. When specified during the inspection sequence, a quantity of boosters shall be sampled from the inspection lot in accordance with MIL-STD-414, Inspection Level IV, Standard Deviation Method, Variability Unknown. The remainder of the lot shall be held in bond pending acceptance of the sample. If the sample does not cause lot rejection (see 4.3.1.3b), the remainder of the lot shall continue through the inspection sequence as shown on Figure 5. Upon completion of the inspection sequence, the remains of the sample boosters shall be disposed of at the discretion of the cognizant demolition officer.

4.3.1.3 Inspection lot acceptance and rejection criteria. Each booster of the inspection lot shall meet all the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections shown on Figure 5, inspection lot acceptance or rejection shall be based on the following criteria:

a. For the inspections above the dashed line where 100-percent inspection is required, acceptance or rejection shall be on an individual basis. The failure of any booster to meet the applicable requirements of Sections 3 and 5 shall cause rejection of the booster.

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b. For the inspections below the dashed line where sampling is required, each sample booster shall meet the applicable requirements of Section 3. For the function and output test of 4.6.1, a depth of dent of less than 0.140 inch from any sample booster shall be cause for rejection of the inspection lot. When computed in accordance with 4.6.1.3d, the inspection lot shall be accepted if the quantity (X-L)/s is equal to or greater than MIL-STD-414, Table B-1, k value, AQL of 0.65 percent defective. A quantity less than the k value shall be cause for rejection of the inspection lot.

4.4 Measuring and test equipment. All measuring and test equipment utilized by the contractor for performance of the inspections specified herein shall be calibrated and maintained in accordance with MIL-STD-45662. When specified in the contract or purchase order (see 6.2.2), a calibration system description shall be made available for review and acceptance by the contracting activity prior to use.

4.4.1 Standard test equipment. The contractor shall provide and maintain inspection and test equipment of sufficient quality and quantity to permit the performance of the required tests and measurements under the conditions and to the accuracies specified herein. Unless otherwise specified herein, the magnitude of any error introduced by the test equipment shall not exceed 10 percent of the allowable tolerance of the requirement being measured.

4.4.2 Special test equipment and circuits. When special test equipment and circuits, other than as specified herein, are devised or when commercially-available equipment is employed, all test equipment, circuits, and methods shall meet the requirements of 4.4 and 4.4.1, and shall be subject to approval by the contracting activity.

4.4.3 Gages. The contractor shall provide whatever gages are necessary to ensure conformance with the dimensional requirements of this specification and the applicable drawings. When specified in the contract or purchase order (see 6.2.1), the contracting activity will furnish drawings of pertinent Government final inspection gages. This procedure shall not relieve the contractor of his responsibility for the design and manufacture of the gages required for satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gages.

4.5 Test conditions and provisions. The following test conditions and provisions shall apply during performance of the inspection methods of 4.6:

a. Test environment. Unless otherwise specified herein or on the drawings, the test environment shall conform to 3.4.3.2.

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b. Test plan. When specified in the contract or purchase order (see 6.2.2), the contractor shall prepare a test plan in accordance with MIL-T-18303, which shall be made available to the contracting activity for review and acceptance prior to use. The contractor shall not deviate from the accepted test plan unless authorized by the contracting activity.

c. Inspection record. When specified in the contract or purchase order (see 6.2.2), the contractor shall prepare an inspection record for each booster produced under a specified contract. The results of all inspections specified herein, which may be extracted from the inspection and acceptance records maintained by the contractor, shall be recorded on the inspection record and shall be traceable to a specific lot.

d. Inspection sequences. The inspections of 4.6 shall be performed in the sequences shown on Figures 4 and 5, as applicable. When feasible, sequential inspections may be combined provided it can be demonstrated by the contractor that the applicable requirements of Sections 3 and 5 have been individually met. The contractor shall not deviate from the inspection methods of 4.6 and additional tests shall not be added unless authorized by the contracting activity.

4.6 Inspection methods. The following inspections shall be performed in accordance with the test conditions and provisions of 4.5.

WARNING

The booster contains explosive materials which are sensitive and hazardous. Safety precautions shall be observed during handling and testing of the booster (see 3.4.2).

4.6.1 Function and output. The function and output test shall consist of initiating the booster on a steel dent block (see 4.6.1.1.1d) and measuring the depth of the dent to determine the output. The test operation and selection of test equipment and materials, except for those listed in 4.6.1.1.1, shall be the responsibility of the contractor. A list of test equipment and materials required for safe performance is given in 4.6.1.1.2 to facilitate test setup by the contractor. To determine conformance to 3.2.1, the booster shall be tested in accordance with 4.6.1.2, the depth of the dent shall be measured and computation shall be performed as specified in 4.6.1.3d.

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4.6.1.1 Function and output test equipment and materials.

4.6.1.1.1 Required test equipment and materials. The following test equipment and materials shall be required to perform the function and output test:

a. Lead holder shown on Figure 2.

b. MARK 9 MOD 0 Explosive Lead in accordance with LD 480141.

c. EBW (see 4.6.1.1.2a).

NOTE: A contractor-selected detonator may be used subject to the approval of the output characteristics by the contracting activity. Approval of the contractor-selected detonator does not imply approval of the detonator's safety characteristics.

d. Dent block shown on Figure 3.

e. Expendable support blocks shown on Figure 3 to sustain the dent block of d. above during the measurement procedure of 4.6.1.3.

f. Dial indicator height gage and probe as specified in MIL-STD-331, Test 301.

g. Detonator firing source, low current resistance measurement device with the current limited to not greater than 10 milliamperes (mA), and interconnecting cables for the detonator of c. above.

4.6.1.1.2 Safe performance test equipment and materials. The following test equipment and materials are required for safe performance of the function and output test in conjunction with suitable safety procedures and facilities:

a. RP-87 EBW detonator.

b. Firing source and cables for the detonator of 4.6.1.1.2a.

c. Firing bay or barricades for the noise, blast and fragment hazards of the booster.

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4.6.1.2 Function and output test method. The booster shall be tested as specified below using the test setup shown on Figure 1:

WARNING

The function and output test involves hazardous operations and shall require the exercise of extremely stringent safety precautions and procedures, experienced ordnance personnel, equipment and facilities, and comprehensive safety reviews (see 3.4.2).

a. Place the steel dent block shown on Figure 3 on the steel plate, as shown on Figure 1.

b. Place the booster in the center of the dent block, booster cap up.

c. Place the loaded lead holder, without the detonator, on the booster as shown on Figure 1.

NOTE

(1) The installation and wiring of the detonator shall be performed in accordance with the contractor's safety procedures (see step 4.6.1.2h).

(2) Steps d. through l. below shall be required as good safety procedures appropriate for the hazards involved.

d. Clear the firing area of all nonessential personnel and equipment.

e. Verify that the detonator leads are shorted and that ordnance personnel, the firing cables, and the booster assembly are grounded.

f. Verify that the firing cables are not connected to the firing source.

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g. Insert the detonator into a containment block, remove the short from the detonator leads, connect the detonator leads to the firing cable, and insulate the connections.

h. Insert the detonator into the lead holder so the detonator is bottomed against the explosive lead, as shown on Figure 1.

i. Remove all remaining personnel and nonexpendable equipment from the test bay and verify that all openings through which fragments could escape are secured.

WARNING

Detonation of the booster causes a significant blast and fragment hazard. Verify that the area is cleared of all personnel before applying energy to the firing circuit (see 3.4.2).

j. Connect the firing cable to the firing source.

k. Verify that all personnel are clear, sound the firing alarm, actuate the red warning light, and initiate the detonator.

l. Allow sufficient time to elapse for the fumes to clear the firing bay, resecure the firing source, disconnect and short the firing cables, and retrieve the dent block.

4.6.1.3 Output dent measurement procedure. The depth of dent shall be measured as follows:

a. Place the dent block on the support blocks shown on Figure 3.

b. As shown on Figure 3, place the gage probe of the dial-indicator height gage (see 4.6.1.1.1f) in the deepest part of the dent within the shaded area. Zero the gage indicator.

NOTE: Do not change the relative position of the dent block on the support blocks after zeroing the gage indicator.

c. Remove the probe from the dent and take gage readings at two points, each approximately 0.125 ±0.030 inch from the opposite edge of the dent block, as shown on Figure 3. Average the two readings to determine the depth of dent and conformance to 3.2.1 (see 4.2.2 and 4.3.1.3b).

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d. Using the data from this test, compute the quantity (X-L)/s, as defined in MIL-STD-414, Single Specification Limit, Form 1, Variability Unknown, Standard Deviation Method. The lower limit, L, of the booster output to be used in the computation shall be 0.145 inch (see 4.2.2 and 4.3.1.3b).

4.6.2 Radiographic. To determine conformance to 3.2.2, a radiographic inspection in accordance with MIL-STD-453 shall be performed on each booster using radiographic equipment capable of detecting assembly defects such as cracks, voids, fissures, low density areas, or loose powder. No less than two X-rays shall be taken of each booster, each 90 ±5 degrees apart and showing the longitudinal section through the booster (see PBL 1137AS2190).

4.6.3 Environmental tests. After exposure to the environments specified in 4.6.3.1 through 4.6.3.3 the booster shall be subjected to the test of 4.6.1 in the sequence shown on Figure 4 or 5, as applicable.

4.6.3.1 Transportation vibration-temperature. To determine conformance to 3.2.3.1, the booster shall be tested as follows:

a. Mount the vibration fixture of Drawing SA 2747612 on a vibration table and tighten the hold-down bolts to a torque of 20 ±1 foot-pounds (ft-lb).

b. Mount the booster to the vibration fixture and tighten the mounting screws to a torque of 17 ±1 inch-pounds (in-lb).

c. Subject the booster to the test of MIL-STD-331, Test 119, Procedure 2, except first article boosters shall be temperature-conditioned to and vibrated at the following temperatures (see Figure 4):

(1) Ten boosters at +160 ±5 degrees Fahrenheit (°F).

(2) Ten boosters at -65 ±5°F.

4.6.3.2 Temperature and humidity. To determine conformance to 3.2.3.2, the booster shall be tested in accordance with MIL-STD-331, Test 105, Two Chamber Method, except the booster shall be exposed to a low temperature of -65°F in Step 5.1.8.

4.6.3.3 Random vibration. The booster shall be tested as follows:

a. Repeat 4.6.3.1a and b.

b. Subject the booster to the test of MIL-STD-810, Method 514.3, Table 514.3-I, Category 7B, Procedure 1, Table 514.3VI, using parametric values f_1 = 1000 Hz, f_2 = 2000 Hz, w_1 = .01 g^2/Hz, w_2 = .12 g^2/Hz. The vibration shall be applied for a period

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of one hour along each of three mutually perpendicular axes, for a total test time of three hours. First article booster shall be temperature-conditioned to and vibrated at the following temperatures (see Figure 4):

(1) Ten boosters at +160 ±5°F.

(2) Ten boosters at -65 ±5°F.

NOTE: To prevent damage to test equipment, limiting of instantaneous acceleration peaks to three times the overall root-mean-square acceleration level is permitted.

The booster shall meet the requirements of 3.2.3.3.

4.6.4 Safety tests.

4.6.4.1 Twelve-meter drop. The booster shall be tested as follows:

a. Mount the booster to the drop test fixture of Drawing SA 2747614 and tighten the hold-down screws to a torque of 17 ±1 in-lb.

b. Subject the booster to the test of MIL-STD-331, Test 103, Procedure 1.

NOTE: Breakage of the explosive pellet and the plastic cup, or the presence of loose pieces of explosive in the drop test fixture shall not be considered criteria for failing the test.

The booster shall meet the requirements of 3.2.4.1.

4.6.4.2 Jolt. To determine conformance to 3.2.4.2, the booster shall be tested as follows:

a. Mount the jolt fixture of Drawing SA 2747613 to the jolt machine and tighten to a torque that will prevent loosening during the test.

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b. Mount the booster to the jolt fixture and tighten the hold-down screws to a torque of 17 ±1 in-lb.

c. Subject the booster to the test of MIL-STD-331, Test 101, Procedure II.

4.6.4.3 Cook-off. To determine conformance to 3.2.4.3, the booster shall be tested as follows:

NOTE: A new thermal insulating cap (67A84D518) will be required for each cook-off test.

WARNING

The cook-off test involves hazardous operations. Safety precautions shall be observed (see 3.4.2).

a. Install the booster in the cook-off test fixture of Drawing SA 1137AS2195 as shown on the drawing and as shown on Figure 6.

NOTE: Use of a high temperature anti-seize compound between all threaded surfaces will greatly facilitate disassembly after the test.

b. Position the assembly of a. above over a gas burner array as shown on Figure 7. The height of the assembly above the array shall be such that the assembly will be engulfed in flame.

NOTE: Unless otherwise specified in the contract or purchase order (see 6.2.1), the test setup of b. above shall be approved by the contracting activity prior to test.

WARNING

During the direct flame impingement of 4.6.4.3c, some portions of the cook-off test fixture thermal insulting cap (see Figure 6) may be expelled violently.

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c. Subject the assembly to direct flame impingement at an average flame temperature of 1,500°F until all ordnance reactions are completed or until no greater than 15 minutes have elapsed. Continuously monitor and record the temperature of the flame at the two locations shown on Figure 7.

4.6.5 Visual. The booster shall be visually inspected to verify conformance to 3.4.1, 3.4.2, and 3.4.3.

4.6.6 Packing. To determine conformance to Section 5, examine the packaging, packing, and marking of the booster prior to shipment.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with FED-STD-102, Level C, and as specified in 5.1.1.

5.1.1 Level C packaging. Boosters shall be packaged separately and in a manner that will provide adequate safety and protection against corrosion, deterioration, and physical damage during direct shipment from the supply source to the first receiving activity. Packaging shall be suitable for explosives and shall conform to 49 CFR 171-178.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level C, and as specified in 5.2.1.

5.2.1 Level C packing. Boosters, packaged in accordance with 5.1, shall be packed in a manner that will provide adequate safety and protection from damage during direct shipment from the supply source to the first receiving activity. Packing shall be suitable for explosives and shall conform to 49 CFR 171-178.

5.3 Marking. When specified in the contract or purchase order (see 6.2.1), all marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The booster is intended for use in the MK 82 warhead with the MK 334 fuze.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

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b. Government-furnished property if other than as specified in 3.1.1.

c. Responsibility for inspection and inspection facilities, if other than as specified in 4.1.

d. Inspection lot if other other than as specified in 4.3.1.

e. Inspection lot size (see 4.3.1.1).

f. Government final inspection gage drawings, if required (see 4.4.3).

g. Approval by contracting activity of cook-off test setup, if required (see Note, 4.6.4.3b).

h. Special marking, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Options
4.4	Calibration System Description	DI-R-7064	---
4.5b	Test Plan	DI-T-5204	---
4.5c	Inspection Record	UDI-M-20404	---

(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DOD 5010.12-L AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.1.1.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of 30 boosters, as specified in 4.2. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

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6.4 Definition of safe. Safe is defined as the ability of the booster to maintain its safety features in a condition which will not create a hazard for personnel or cause any subsequent action to occur during its service life which will nullify the required safety conditions of service use. For nonservice use, safe is defined as the ability of the booster to maintain its safety features in a condition which will permit safe disposal without injury to personnel.

6.5 Subject term (key word) listing.

Booster, Fuze
Fuze (ordnance)
Missile, guided

Preparing activity:
Navy - AS
(Project 1336-N570)

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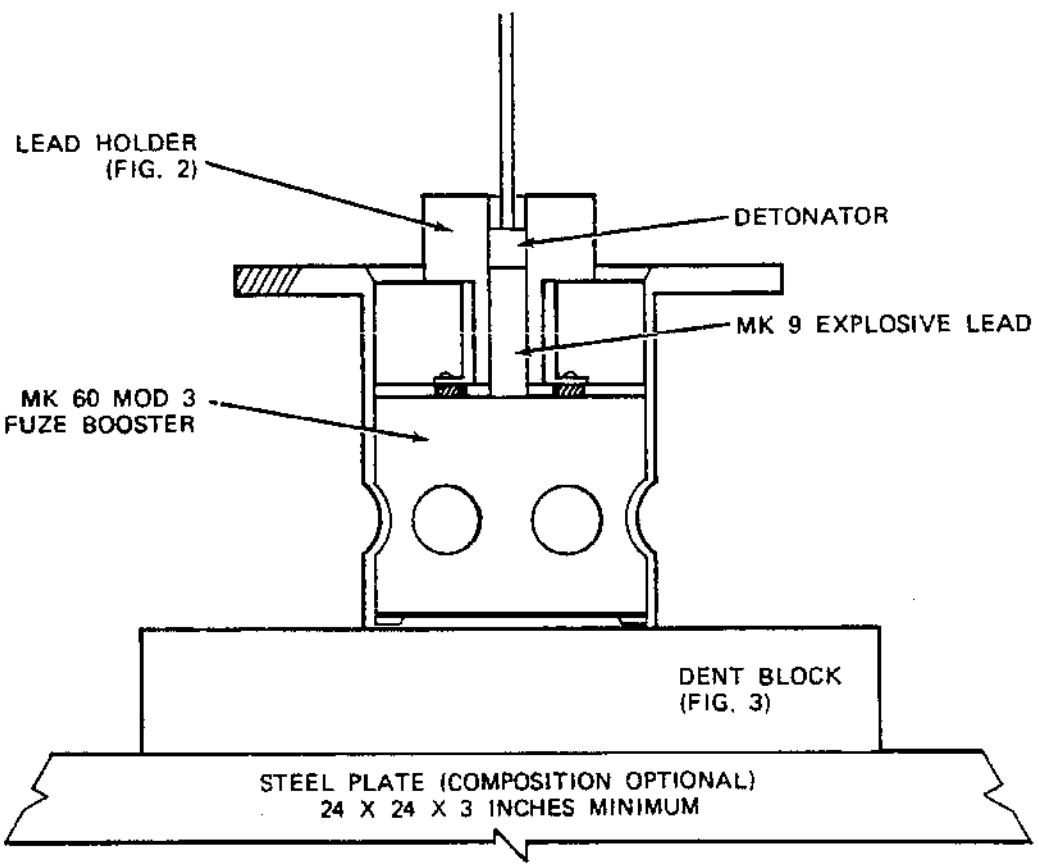
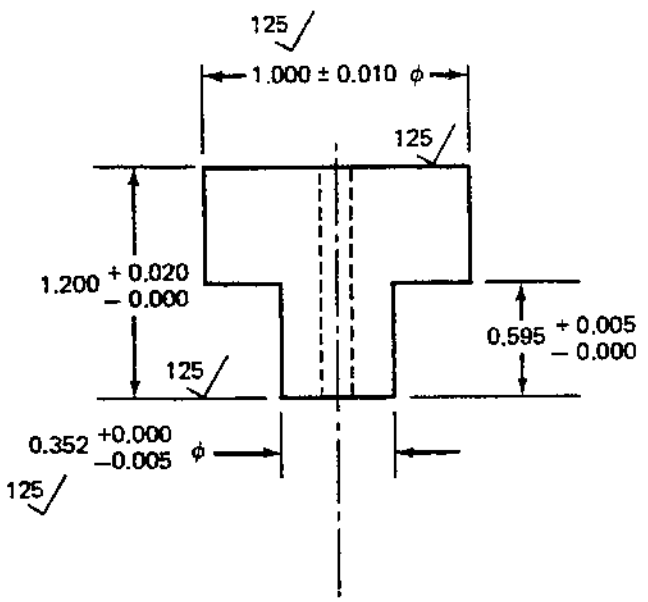


FIGURE 1. Function and output test setup.

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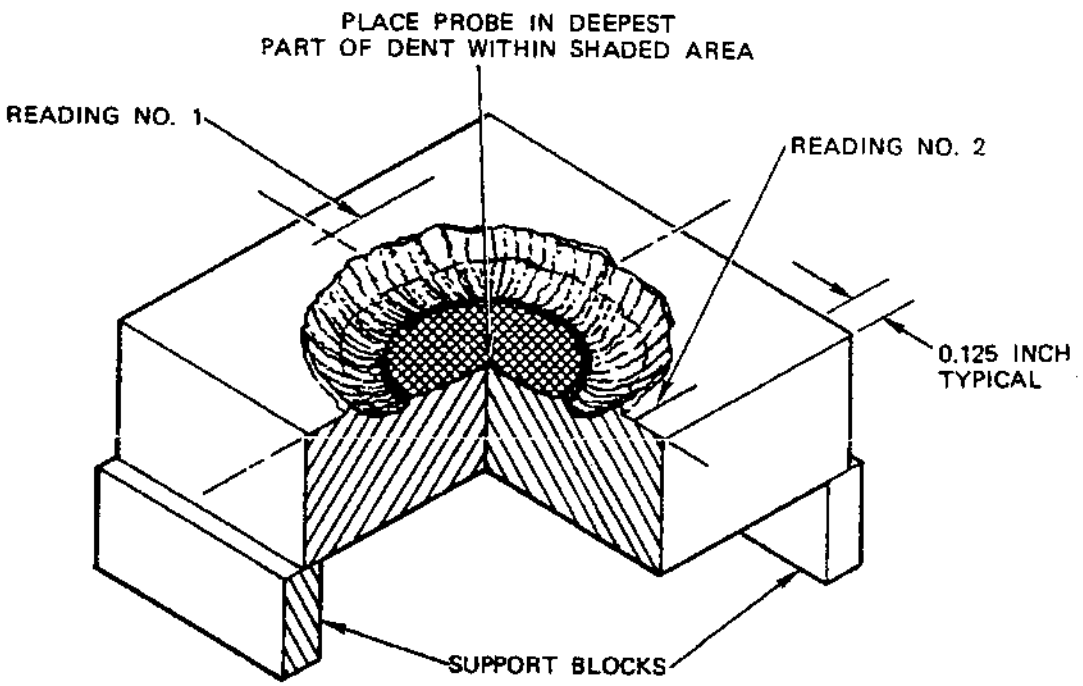


NOTES:

1. Material shall be steel, ASTM A108 Grade 1018 through 1035, Condition optional.
2. Dimensions are in inches.

FIGURE 2. Lead holder for function and output test.

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NOTES:

1. Dent Block: Steel, cold finished bar in accordance with ASTM A108, Grade 1018 or 1020, Rockwell hardness B70 to B95, surface condition as rolled. Size 5.000 ±0.125 inches square, 1.5 inch stock. Upper and lower surfaces of block shall be clean and free from rust and burrs.
2. Support Blocks: Material and size shall be at the contractor's discretion.

FIGURE 3. Dent block and output dent measurement for function and output test.

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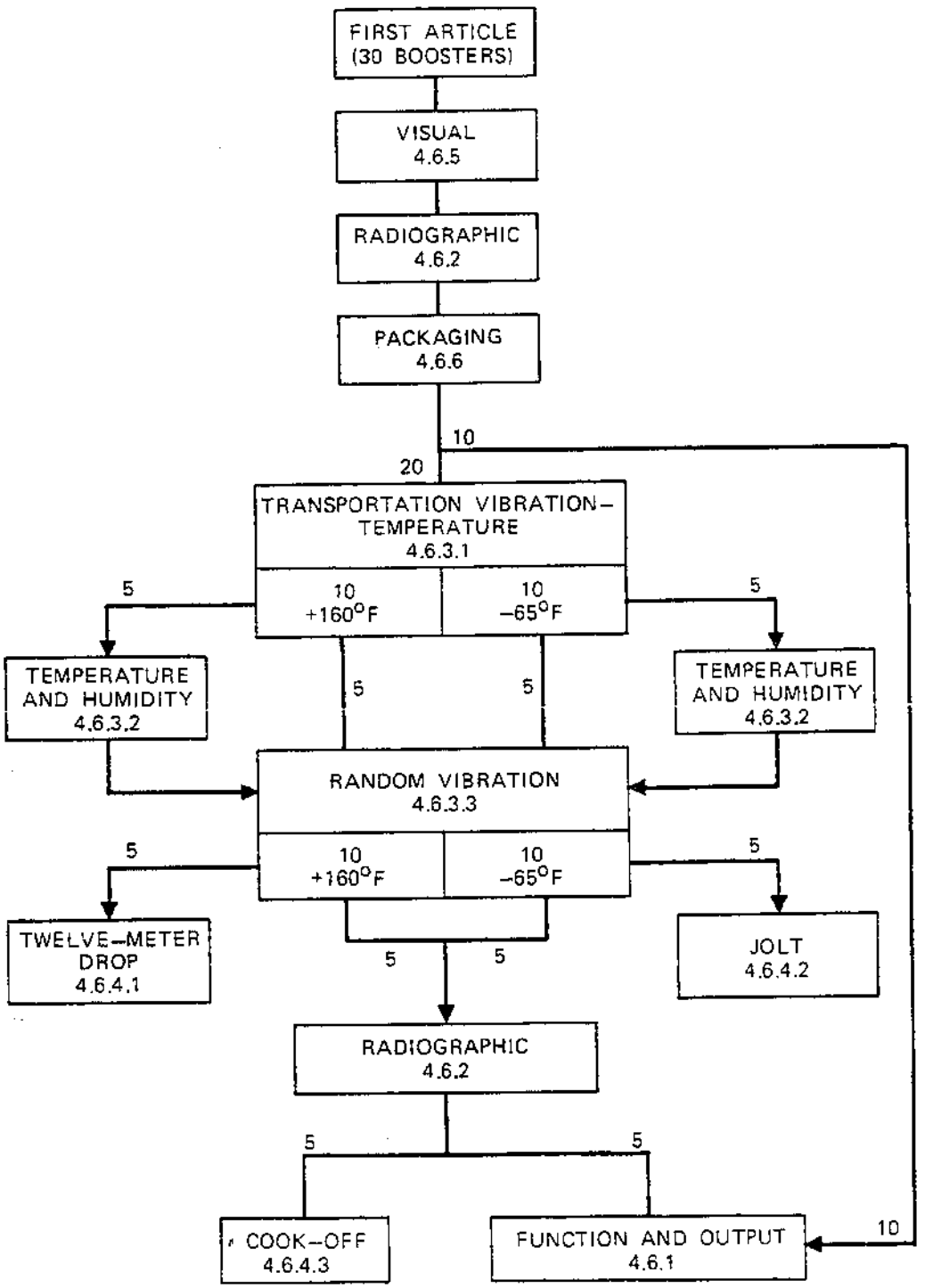


FIGURE 4. First article acceptance inspection sequence.

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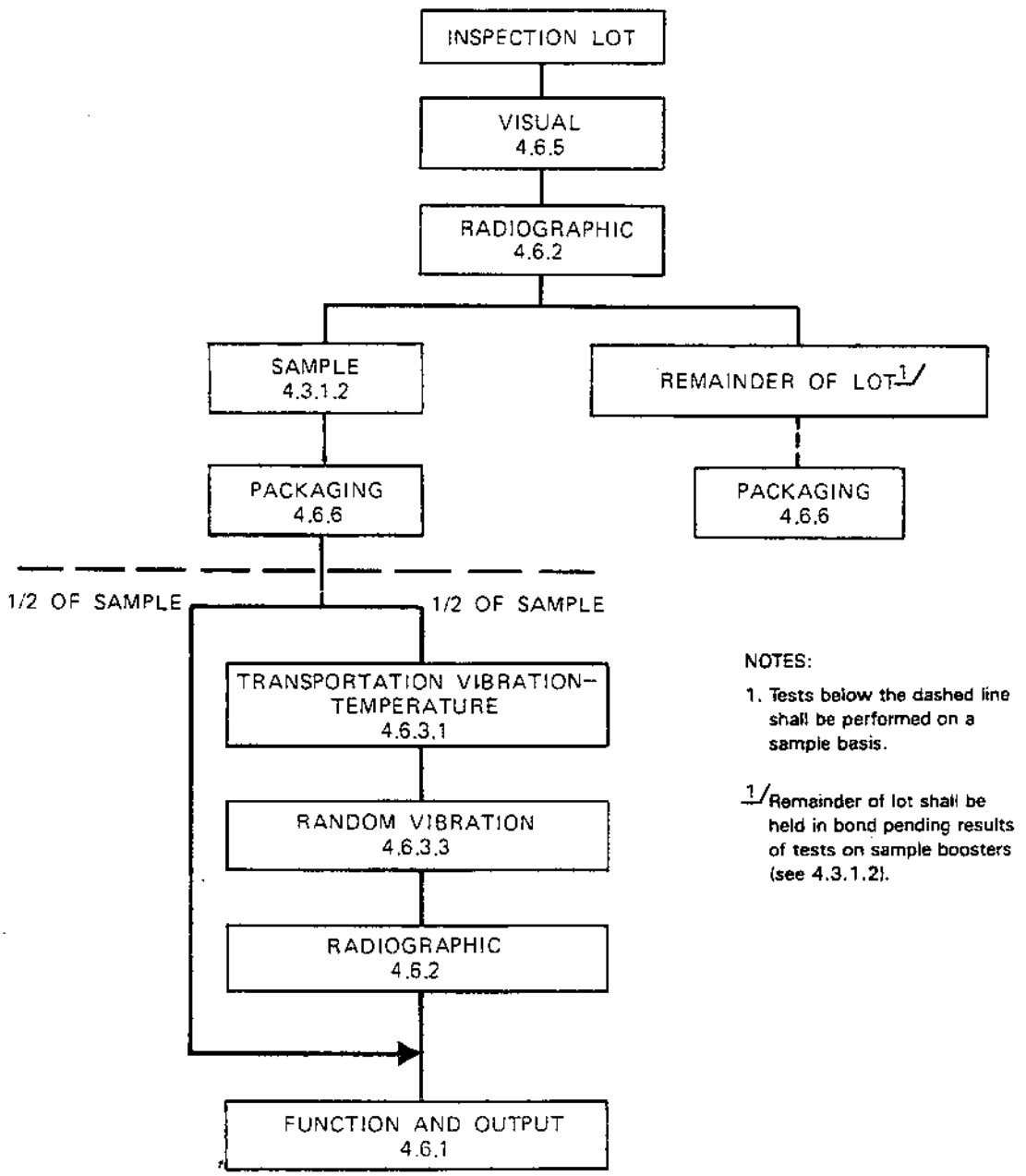
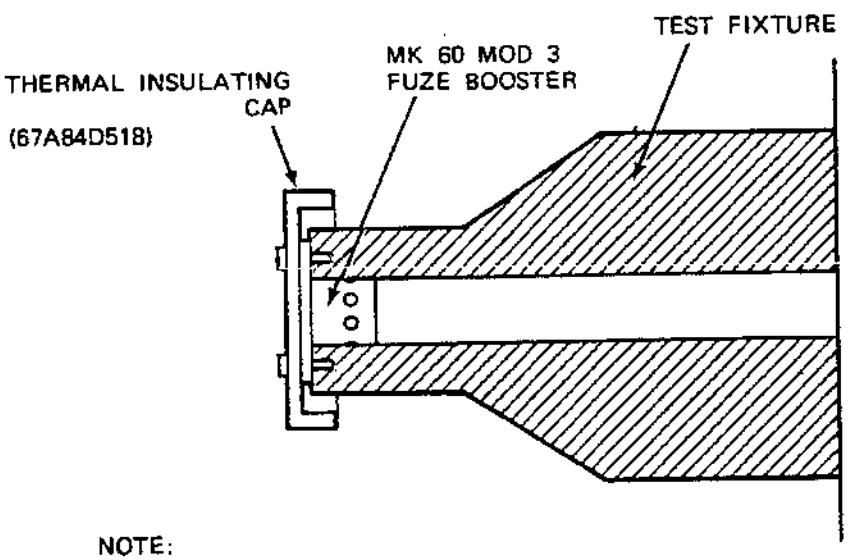


FIGURE 5. Inspection lot acceptance inspection sequence.

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NOTE:

1. Fixture is in accordance with Drawing SA1137AS2195. Figure is designed to simulate the thermal response of the MK 82 MOD 0 Warhead.

FIGURE 6. Booster installed in test fixture for cook-off test.

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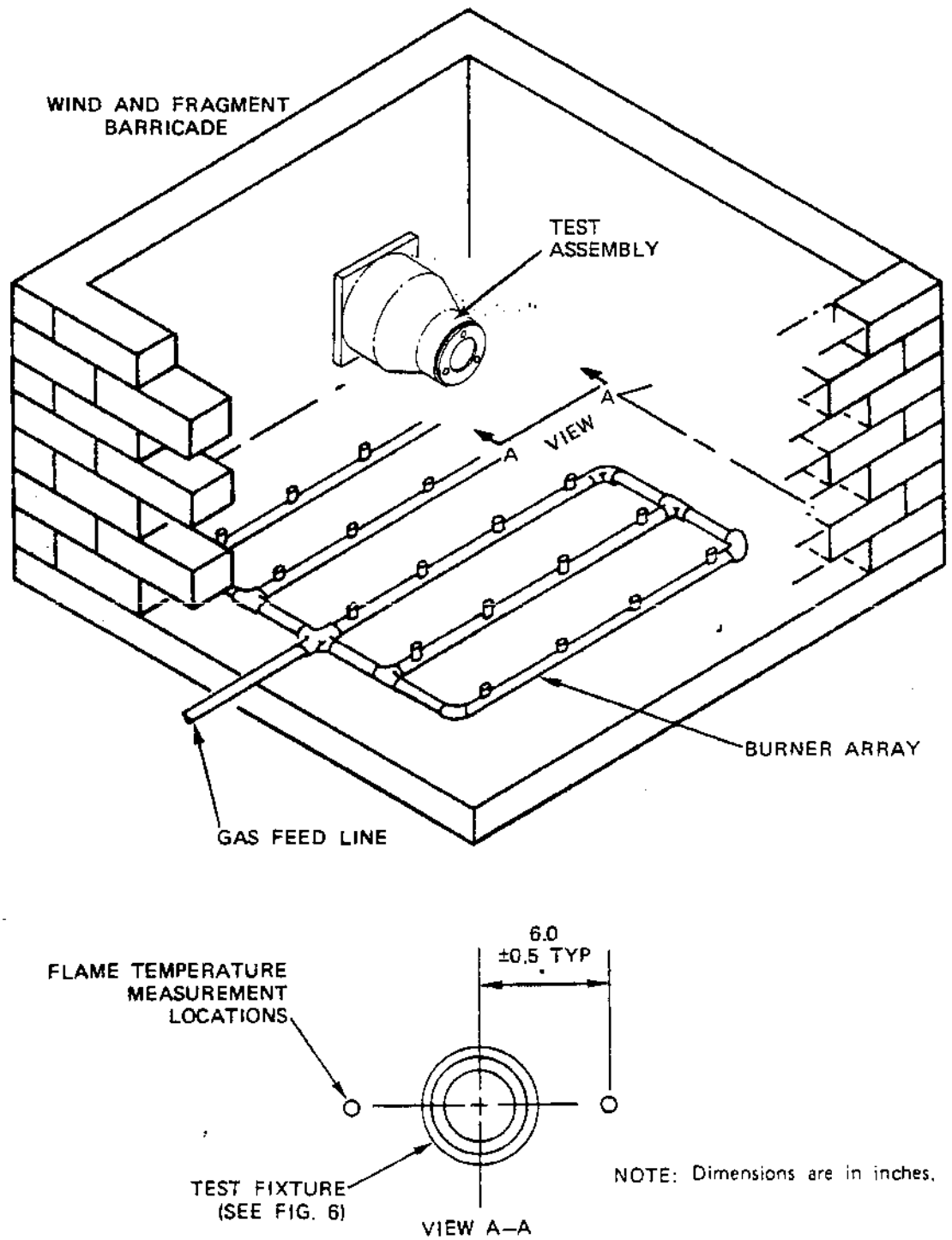


FIGURE 7. Cook-off test setup.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-85685(AS)	BOOSTER, FUZE, MARK 60 MOD 3

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include A Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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